Consistency Is the Only Signal That Transfers: A Preregistered Three-Lineage Map of Cheap Uncertainty Signals in Small Instruct Models

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Abstract

Practitioners routinely estimate whether a language model's answer is correct from cheap, inference-time signals: verbalized confidence, token log-probabilities, token-level entropy, or agreement across resampled generations. We preregistered five predictions about what these signals measure and tested them on a frozen protocol—13 standard benchmarks across five domains, $\sim$200 items each, identical prompts, one shared extraction module—in three independent 3–4B instruct lineages (Qwen2.5-3B, Llama-3.2-3B, Gemma-3-4B). Most predictions failed, and the failures organize into a sharper map. (1) K-sample answer consistency (K=5) discriminates correctness at AUROC $.740/.746/.756$ across the three lineages—the only signal whose performance transfers. (2) Every token-level signal is lineage-specific: mean log-probability ranges from mildly useful ($.61$) to below chance ($.39$); token entropy from $.61$ to chance. (3) Verbalized confidence has three phenotypes—collapsed ($\sim 96\%$ "100", robust to four elicitation wordings including an inverted scale), graded-informative ($.615$), and compressed-high ($.605$). (4) An information decomposition explains the structure: $I(\text{consistency}; \text{correctness} \mid \text{entropy})$ exceeds its shuffled-control floor in $12/12$ model$\times$domain cells, while $I(\text{entropy}; \text{correctness} \mid \text{consistency})$ does so in only $2/12$— uncertainty information lives in the distribution over *outcomes*, not over *tokens*; the two entropies are nearly uncorrelated ($\rho = -.05$ to $.31$). (5) Mechanistically, consistency behaves like a measure of generation-basin depth: single-path agreement is more reliable than divergent-path agreement ($.87$–$.88$ vs $.70$–$.78$), and its failure mode is a distinct population of convergent misconceptions (6–10% of items, precision ceiling $\approx .85$–$.88$ at full agreement) concentrated on plausibility-shaped benchmarks.

1 Introduction

When a deployed language model answers a question, the deployer's first practical question is: *should I trust this one?* Full solutions—training probes, ensembles of models, retrieval verification—are often unavailable at inference time. What remain are cheap signals computable from the model's own generation: ask the model how confident it is; read the log-probability or entropy of the tokens it produced; or sample K additional generations and count agreement.

These signals are used interchangeably as "model uncertainty," and the literature evaluates them mostly one model family at a time. Whether a signal validated on one lineage means the same thing on another is rarely tested, and the question of what each signal *measures*— as opposed to how well it scores on one benchmark—is usually left implicit.

We approached this as a measurement problem with a preregistered design. Before data collection we froze: five falsifiable predictions (P1–P5) inherited from a pilot study on a custom benchmark; a 13-benchmark, five-domain evaluation suite with per-item frozen prompts; a single measurement protocol (greedy chain-of-thought pass, K=5 resampling at temperature 0.7, one fixed confidence elicitation); exclusion rules (accuracy outside $[.10, .90]$; extraction failure $>10\%$ per cell); and the statistical machinery (bootstrap CIs on everything, permutation tests with Holm correction for the primary family, shuffled-control floors under every plug-in information estimate). We then ran the frozen protocol on three independent model lineages at matched scale.

Contributions.

1. A preregistered, QC-disciplined map of four cheap uncertainty-signal families $\times$ 13 benchmarks $\times$ 3 model lineages (7,620 measured items), with all raw rows, code, and generated tables released.

2. The finding that **only behavioral consistency transfers across lineages** (AUROC .74–.76 in all three), while token-level signals vary from useful to anti-predictive and verbalized confidence splits into three lineage phenotypes—one of which (total collapse) we show is robust to four elicitation wordings.
3. An information-theoretic decomposition establishing that consistency contains correctness information that token entropy does not ($12/12$ cells), while the converse fails ($10/12$)—i.e., "uncertainty" factorizes into an outcome-level quantity that tracks correctness and a token-level quantity that mostly does not.
4. A mechanistic characterization: consistency reads the depth of the generation basin (single-path agreement $>$ divergent-path agreement), and its residual errors are convergent misconceptions—a structured, benchmark-localized population, not noise.

2 Related work

Verbalized confidence. Language models can be prompted to state confidence in words or numbers (Lin et al., 2022a; Kadavath et al., 2022; Tian et al., 2023). Findings there are dominated by larger models; our results show that at 3–4B the very *existence* of a graded verbal signal is lineage-dependent.

Self-consistency and resampling. Wang et al. (2023) introduced majority voting over sampled chains to improve accuracy; subsequent work uses agreement as a confidence proxy. We measure what that proxy contains, relative to and conditional on other signals.

Semantic entropy and generation-level uncertainty. Kuhn et al. (2023) and Farquhar et al. (2024) argue for uncertainty measured over meanings rather than tokens. Our answer-distribution/token-distribution dissociation (ρ down to $-.05$; AUROC .70 vs .49 in the same model) is a within-protocol, cross-lineage confirmation of that program's central premise, obtained with exact-match answer equivalence rather than entailment clustering.

Calibration of neural networks. Guo et al. (2017) established the modern calibration toolkit (ECE, temperature scaling); we report quantile-binned ECE/Brier alongside discrimination and keep the two questions separate.

Benchmarks. MMLU (Hendrycks et al., 2021), MMLU-Pro (Wang et al., 2024), TruthfulQA (Lin et al., 2022b), GSM8K (Cobbe et al., 2021), SVAMP (Patel et al., 2021), MultiArith (Roy and Roth, 2015), CommonsenseQA (Talmor et al., 2019), ARC-Challenge (Clark et al., 2018), StrategyQA (Geva et al., 2021), BIG-Bench Hard (Suzgun et al., 2022), MuSR (Sprague et al., 2024), HumanEval (Chen et al., 2021), MBPP (Austin et al., 2021).

Information decomposition. Conditional mutual information and the Williams–Beer partial information decomposition (Williams and Beer, 2010) provide the machinery for the unique-information claims; we guard the plug-in estimators with shuffled-control floors throughout.

3 Methods

Models. Qwen2.5-3B-Instruct, Llama-3.2-3B-Instruct, Gemma-3-4B-IT—three independent pretraining/post-training lineages at matched scale, run fp16 on T4 GPUs with each model's own chat template and no system prompt (byte-identical user prompts).[1]

Benchmarks and domains. 13 benchmarks spanning knowledge (MMLU, MMLU-Pro, TruthfulQA), arithmetic (GSM8K, SVAMP, MultiArith), commonsense (CommonsenseQA, ARC-Challenge, StrategyQA), reasoning (BBH 6-task subset, MuSR), and coding (HumanEval, MBPP). n=200 per benchmark (or full split if smaller; 2,540 items/model), sampled once with seed 0 and frozen to versioned JSONL with a manifest before any model ran.

[1] Camera-ready should cite the three model technical reports; omitted here pending per-venue citation preferences.

Table 1: Mean AUROC (correct vs. incorrect) over unsaturated, extraction-clean cells. Consistency is lineage-stable and best everywhere; token-level signals vary across lineages, and Gemma's mean log-probability is below chance.

signal	Qwen2.5-3B	Llama-3.2-3B	Gemma-3-4B
consistency (K=5)	**.740**	**.746**	**.756**
verbal confidence	.519	.615	.605
token entropy	.610	.614	.544
mean log-probability	.606	.606	**.391**

Protocol (frozen). Per item: (1) greedy CoT pass (512 new tokens max) with token-level statistics—mean log-probability and mean entropy over the full chain and over the answer span, top-1/top-2 margin, top-5 mass; (2) K=5 sampled passes (T=0.7, top-p 0.9) $\rightarrow$ consistency = modal-answer fraction (code: output-agreement on prompt-visible examples) and the answer-distribution entropy; (3) a fixed follow-up turn eliciting confidence 0–100. One shared extraction/grading module across models (multiple-choice letter, numeric, yes/no, executed code); unparseable answers are graded wrong and *flagged*, never dropped silently.

Preregistered predictions. P1: verbal confidence discriminates on knowledge but not arithmetic (ΔAUROC $\geq .15$). P2: consistency's discrimination transfers across domains better than confidence's. P3: full-chain log-probability is weak on arithmetic (fluency, not correctness). P4: consistency and answer-entropy are the same signal ($\rho > .95$ everywhere; falsified by any benchmark $< .9$). P5: a zero-cost input feature rivals internal signals where difficulty is input-legible, and no input feature approaches confidence on knowledge.

QC and statistics. Saturated cells (accuracy outside $[.10, .90]$) and cells with $>10\%$ extraction failure are excluded from primary contrasts; both exclusions have flagged-included sensitivity variants. 95% percentile bootstrap CIs on every statistic; permutation tests (Holm-corrected within the primary family) for domain contrasts; every MI/CMI/PID estimate is reported against the same estimator run on label-preserving shuffled signals. Wording sensitivity of the confidence elicitation was probed post hoc under preregistration cover (three alternative phrasings, M1, MMLU+GSM8K, n=100): probability framing, frequency framing, and an inverted scale.

Independent audit. Every headline and secondary number below was recomputed by a second implementation sharing no code with the analysis pipeline (data integrity via SHA256, the signal map, P4, the information asymmetry, phenotypes, QC rates, and mechanism), reading only the frozen rows. All reproduce; details in the supplement.

4 Results

4.1 The signal map

Table 1 gives mean AUROC over unsaturated, extraction-clean cells. Consistency is the best signal in essentially every benchmark$\times$model cell and its level is lineage-stable. Token-level signals are not: Gemma's mean log-probability is anti-predictive (below chance) on clean cells—fluent chains negatively associated with correctness—making Gemma the only lineage where preregistered P3 was supported (arithmetic $\text{logprob}_{\text{full}} = .297$). Verbal confidence spans three phenotypes: Qwen answers "100" on 95.6% of items (mean confidence $.992$ when correct, $.954$ when wrong); Llama produces a graded distribution (modal 80) at $.615$; Gemma a compressed-high distribution (67% at "95") at $.605$.

4.2 Preregistered verdicts

Under strict QC, P1 is falsified in Llama (ΔAUROC $= -.083$, 95% CI $[-.152, -.019]$— *sign-reversed*: arithmetic confidence discriminates better than knowledge) and Gemma ($-.124$, $[-.192, -.052]$), and not evaluable in Qwen (the arithmetic pool empties after exclusions; the labelled sensitivity variant gives $+.008$ $[-.040, +.040]$, falsified). P2 is supported in Qwen and Llama, falsified in Gemma. P4 is falsified in all three lineages (min $\rho = .001/.009/-.197$ against a predicted $> .95$). P5's preregistered falsifiers fire in all three (Table 2).

Table 2: Preregistered predictions across the three lineages. No prediction survives in all three; P4 fails identically in all three, and its failure mode is the central finding.

	Qwen2.5-3B	Llama-3.2-3B	Gemma-3-4B
P1 domain gap ($\geq .15$)	inconclusive (QC); sens. falsified $\Delta{=}+.008$	**falsified** $\Delta{=}-.083$ (sign-reversed)	**falsified** $\Delta{=}-.124$
P2 consistency transfers	supported (sens.)	supported	falsified
P3 full-chain logprob weak on arith.	falsified (.725)	falsified (.800)	**supported** (.297)
P4 consistency $\approx$ ans.-entropy	falsified ($\rho_{\min}{\approx}.001$)	falsified (.009)	falsified ($-.197$)
P5 free-feature gate	falsified	falsified	falsified

4.3 Elicitation robustness: the collapse is the model, not the prompt

All four wordings on Qwen produce $\geq 94\%$ modal replies with chance-level discrimination (AUROC .47–.57, every CI spanning .5; parse rate 100%). The decisive cell is the inverted scale: asked how likely its answer is *wrong*, the model answers "0" (94%)— it flips the number with the semantics. The collapse is semantically coherent, genuinely held total certainty, not number-anchoring or elicitation noncompliance.

4.4 Information structure: outcome uncertainty, not token uncertainty

Per model$\times$domain (never pooled), with $C =$ consistency, $E =$ answer-span token entropy, $Y =$ correctness:

- $I(C; Y \mid E)$ exceeds its shuffled-control floor in $\mathbf{12/12}$ cells (.009–.125 nats, typically 10–30$\times$ floor).
- $I(E; Y \mid C)$ does so in $\mathbf{2/12}$ cells, both marginal ($\leq .02$ nats).
- $\rho(C, E)$ ranges $-.20$ to $+.27$ across cells.

The asymmetry—consistency subsumes entropy's correctness information but not conversely— replicates in all three lineages (Fig. 1). A direct comparison of the two "entropies" sharpens it: the entropy of the K-sample *answer distribution* discriminates at .695–.731 in all three models, while *token-level* entropy ranges .494–.706, and the two are nearly uncorrelated ($\rho = -.05/.24/.31$; Fig. 1A,C). Conditioning on the best zero-cost input feature removes consistency's information in only $2/15$ cells, both in the reasoning domain. An independent re-implementation with a multi-permutation bias floor reproduces the asymmetry ($10/12$ vs $0/12$); the two cells where consistency's margin does not clear the stricter floor are both the reasoning domain—the weakest-signal domain. The direction $C \gg E$ is estimator-robust; the exact "$12/12$" should be read as "in every domain except the weakest."

4.5 Mechanism: basin depth and convergent misconceptions

$P(\text{correct})$ is a staircase in consistency ($C \leq .4$: .28–.43; mid: .41–.52; $C \geq .8$: .78–.82) and is flat in entropy within each band. Among high-consistency items, agreement reached along essentially a single reasoning path is more reliable than agreement reached along divergent paths—.87–.88 vs .70–.78 correct, in all three models. The residual failure mode is a distinct population: wrong-but-fully-consistent items are 6.3– 10.4% of all items (precision at $C{=}1$: .84–.88), are 91–97% extraction-clean, sit on prompts 1.5–1.9$\times$ longer than average, and concentrate in TruthfulQA, MuSR, StrategyQA, and CommonsenseQA—benchmarks shaped around plausible misconceptions. Consistency correlates only weakly with generation length ($\rho = -.22$ to $-.34$) and prompt size ($\approx -.1$): it is not a repackaged difficulty feature.

4.6 Quality control

Cells exceeding the preregistered 10% extraction-failure threshold (all excluded from primary contrasts, all with sensitivity variants): Qwen—BBH 13.1%, GSM8K 12.5%, MMLU-Pro 46.5%; Llama—MMLU-Pro 32.0%, MBPP 68.5%; Gemma—MMLU-Pro 53.0%, MuSR 13.1%, StrategyQA 14.5%. Diagnosis (no parser bugs found): truncation at the frozen 512-token budget and answer-format drift in which the preregistered fallback still recovered a graded answer.

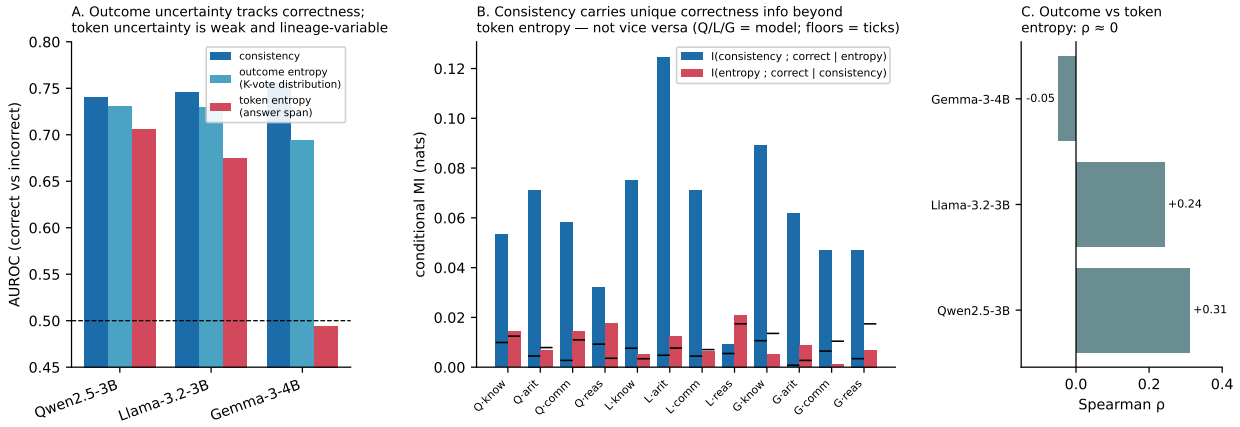


Figure 1: **Entropy is not one quantity.** (A) Consistency and the K-vote *outcome*-entropy track correctness (.70–.76); answer-span *token*-entropy is weaker and lineage-variable (Gemma below chance). (B) $I(C;Y \mid E)$ (blue) vs $I(E;Y \mid C)$ (red) per model$\times$domain, with shuffled-control floors as black ticks: consistency carries unique correctness information beyond token entropy—not vice versa. (C) Spearman ρ between outcome- and token-entropy is $\approx$0.

5 Discussion

One signal, three lineages. Of the four cheap signal families, only the K-sample vote means the same thing across model families. Token-level statistics—the cheapest and most-used proxies—change sign across lineages. A token-level uncertainty threshold tuned on one family and shipped on another is an uncalibrated instrument; in the Gemma case it would be systematically wrong.

Why P4 failed. "Entropy" conflated two quantities. Uncertainty over *which tokens to emit next* is dominated by lineage-specific surface variance (phrasing, formatting, chain length); uncertainty over *which answer the process lands on when perturbed* is the correctness-relevant part—and only resampling reads it. This is a within-protocol, exact-match analogue of the semantic-uncertainty argument, demonstrated as a double dissociation inside a single model (Gemma: answer-distribution entropy .695, token entropy .494, $\rho = -.05$).

What consistency measures. The path-divergence result argues that consistency is a probe of *basin depth*: when the model knows, temperature perturbation barely moves the trajectory. That framing predicts the observed failure mode—deep basins encode shared misconceptions as readily as knowledge, so the residual errors of a consistency-gated system are convergent misconceptions on plausibility-shaped inputs. It also explains why divergent paths converging on one answer are *weaker* evidence than a single stable path, inverting the ensemble intuition that independent agreement is stronger.

Verbal confidence. At this scale, whether graded verbal uncertainty exists at all is a training-lineage property (three phenotypes under one protocol), and where it is absent it is absent robustly (four wordings, semantics demonstrably understood). Claims of the form "LLMs can/cannot verbalize uncertainty" need a lineage qualifier.

Practical guidance for 3–4B deployments. Spend the K-sample budget; gate on agreement; expect $\approx$.85 precision at full agreement; and treat the remaining errors as misconception-shaped, not noise. Token statistics should be used only with per-lineage validation, and verbal confidence only after checking the lineage produces one.

6 Limitations

Three lineages at 3–4.3B; findings may not extend to frontier scale. Single seed so far (the preregistered seed run is built but unexecuted; all claims carry that caveat). The frozen 512-token budget truncates verbose models on long-form benchmarks—flagged, excluded from primary contrasts, and diagnosed. $K{=}5$ is small; consistency at .74–.76 is likely a floor. Exact-match/execution grading is conservative. MI/CMI/PID are plug-in estimates guarded by shuffled controls, not debiased estimators. GPQA was inaccessible (gated). Contamination of older benchmarks is plausible; we measure uncertainty–correctness relationships, not capability.

7 Reproducibility

Everything derives from: frozen benchmark JSONLs with manifest; a frozen measurement protocol, audited by a smoke test per model before any full run; raw rows committed per session with an append-only kernel audit trail; a SHA256-manifested master dataset; analyses that read only frozen rows and emit every table/figure cited here; a timestamped deviations log; and an independent second-implementation audit of every reported number. The preregistration predates all data.

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